

Open-Gate Closure Note for Article and Supplement v15

Lean-facing source closure, fail-closed Gate Zero audit, executed direct recurrence, receiver-study package, and external-mapping certificates

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Abstract

Article and Supplement v15 leave five gates explicit: a Lean kernel build, a v15 Gate Zero audit of the inherited finite realization, a direct recurrence experiment, an independent receiver study, and external mappings. This note attempts closure rather than replacing those gates with prose assertions. One gate is positively closed in the declared finite model: the direct recurrence experiment was preregistered in executable form and run for 1,000 common-design replicates in each of five post-lift regimes. One gate is closed negatively: the Gate Zero audit was completed over every artifact available in the current package and returned a fail-closed result because the v8 generator, event log, trajectory table, registry, frozen configuration, conformance suite, environment lock, source figures, and checksums were not present. The Lean package is source-complete at a bounded theorem level and passes a static no-sorry/no-custom-axiom scan, but no Lean executable was available in the environment, so kernel closure is not claimed. The receiver protocol, hidden task bank, answer key, and scoring schema are complete, but recruitment and independent execution remain empirical. External mappings are registration-closed through an evidence-and-defeater matrix, while physical, AI, historical, branch, hazard, and residue instantiation remain open. The result is a complete disposition: every gate is either closed with an executed result or fails closed with a precise remaining input, and no unavailable evidence is silently promoted.

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1 Purpose and governing rule

The purpose of this note is to close the five open v15 gates wherever the available environment and artifacts permit. Closure is not defined as producing a favorable answer. A gate is closed when its registered procedure has been executed and yields a determinate disposition. A negative result can therefore close an audit gate while simultaneously blocking scientific promotion.

Definition 1.1 (Gate dispositions). For a registered gate G , the permitted dispositions are:

1. *closed-positive*: the required procedure was executed and its acceptance criteria were met;
2. *closed-negative*: the required procedure was executed and a preregistered blocker or defeating result was found;
3. *partially closed*: all objects that can be constructed without the missing external input are complete, but the decisive execution has not occurred;
4. *open*: neither the decisive procedure nor a fail-closed audit has been completed.

The v15 anti-promotion rule is retained:

absence of a defeating result $\nRightarrow$ closure, formal closure $\nRightarrow$ external instantiation.

1.1 Result summary

Table 1: Closure dispositions produced by this note.

Gate		Disposition	Result
Lean kernel build		partially closed	A pinned Lean 4.32.0 source package, theorem modules, dependency prints, and a static no-sorry/no-custom-axiom audit were created. No Lean executable was available, so no kernel acceptance is asserted.
Gate Zero v15		closed-negative	The documentary and artifact-presence audit ran. The inherited numerical lineage cannot be regenerated because every critical v8 execution artifact is absent from the available package.
Direct recurrence		closed	The finite stochastic experiment was executed. Fixed post-lift capacity and heterogeneous branches recurrently fragmented; proportional service, regenerative service, and a second lift prevented the registered persistent-recurrence endpoint.
Independent ceiver study	re-	partially closed	The protocol, 28 hidden tasks, separated answer key, scoring schema, blinding rule, and retransmission endpoint are frozen. No independent receivers were recruited or run.
External pings	map-	partially closed	Each mapping now has a formal object, minimum evidence, strongest permitted claim, and defeater. No physical, AI, historical, branch, hazard, or residue mapping is promoted.

The strongest scientifically new result in this package is therefore the direct recurrence result in the declared synthetic queue-and-region model. The strongest carrier result is the negative Gate Zero audit, which requires the inherited v8 numbers to remain historical constructive-witness claims until their source lineage is supplied.

2 Inputs, trust boundaries, and reproducibility package

The closure attempt used Article v15, Supplement v15, the v15 revision and experiment plan, the manuscript-generation scripts available in the working directory, and the companion source package generated with this note. It did not have access to the v8 event-sourced computational repository described by the manuscripts.

The companion archive contains:

- `lean/`: a pinned Lean-facing project with seven modules and top-level axiom-print commands;
- `code/gate_zero_audit.py`: the artifact and documentary audit;
- `code/direct_recurrence.py`: the executed recurrence experiment;
- `data/`: frozen recurrence protocol, summary data, and metrics;
- `receiver/`: receiver protocol, hidden tasks, answer key, and scoring schema;
- `external/`: external-mapping certificate matrix;
- `audit/`: Gate Zero, claim crosswalk, hashes, and Lean static-audit results;
- `package_manifest.json`: file sizes and SHA-256 hashes.

The artifact can be regenerated with `run_all.sh`. That script invokes a Lean kernel only when `lake` is installed. Its absence produces a visible failure message rather than a simulated pass.

3 Lean-facing formal closure

3.1 What was constructed

The source package pins `leanprover/lean4:v4.32.0` and separates the formal object into the modules in Table 2.

Table 2: Lean-facing source modules.

Module	Main declarations
<code>Basic.lean</code>	typed layers and operations; Register, Express, Store, Recall, L , G ; pointed systems and admissible maps
<code>Semantics.lean</code>	semantic-equivalence theorem for invariant targets; finite semantic non-equivalence witness
<code>Projection.lean</code>	first-order conflation; extensional local-stability iterate
<code>Operations.lean</code>	typed L/G word algebra; Register and Recall nonreduction by domain/codomain obstruction
<code>Order.lean</code>	finite certified families; order increment; certified-family order invariance
<code>Backlog.lean</code>	finite unit-gap backlog growth, reversal, and renewed post-lift growth
<code>GreatFilter.lean</code>	a top-level conditional filter theorem exposing every external premise as a parameter

This is deliberately a bounded formal core. It does not pretend to contain the complete quotient construction for semantic isomorphism classes, the full target language, the entire regional-fragmentation graph theorem, or the published dynamical-CNS bridge. It does, however, replace an empty skeleton with proof-bearing source for the primary logical shapes that the manuscript uses.

3.2 Representative formal statements

Theorem 3.1 (First-order conflation). *Let $q : W \rightarrow O$ and $\Phi : W \rightarrow Y$. If $q(x) = q(y)$ and $\Phi(x) \neq \Phi(y)$, then no decoder $d : O \rightarrow Y$ satisfies both $d(q(x)) = \Phi(x)$ and $d(q(y)) = \Phi(y)$.*

Proof. If such d existed, then

$$\Phi(x) = d(q(x)) = d(q(y)) = \Phi(y),$$

contradicting $\Phi(x) \neq \Phi(y)$. The Lean source realizes this equality chain directly. $\square$

Theorem 3.2 (Typed Register/Recall nonreduction). *Every word generated solely by L and G has conceptual domain and conceptual codomain. Register has signal domain and conceptual codomain; Recall has memory domain and conceptual codomain. Therefore neither Register nor Recall is equal to an L/G word unless an additional typed bridge changes the algebra.*

Proof. Inductively, the generated word type is a conceptual endomorphism. Equality of typed operations implies equality of domains. Signal, memory, and conceptual layers are distinct constructors, so either proposed equality yields a constructor contradiction. $\square$

Theorem 3.3 (Finite backlog growth). *If $U : \mathbb{N} \rightarrow \mathbb{N}$ satisfies $U(t + 1) \geq U(t) + 1$ for every t , then $U(n) \geq U(0) + n$ for every n .*

Proof. Induct on n . The successor case combines the induction hypothesis with the one-step gap. The source uses Lean’s Presburger arithmetic tactic after exposing the two inequalities. $\square$

Theorem 3.4 (Renewed growth after a lift). *Fix a post-lift time T . If $U(T + k + 1) \geq U(T + k) + 1$ for every k , then $U(T + n) \geq U(T) + n$ for every n .*

Proof. Apply the preceding induction to the shifted sequence $k \mapsto U(T + k)$. $\square$

3.3 Static source audit

The static audit scanned every `.lean` file. It found:

Forbidden token	Count
sorry	0
admit	0
axiom	0
opaque	0

This result is useful but not a kernel result. Lexical absence of `sorry` does not establish that the files elaborate or that the proof terms type-check.

3.4 Why the kernel gate is not closed

No lean or lake executable was installed in the available execution environment. A release binary could not be materialized into that environment. Consequently, the following acceptance criteria remain unexecuted:

1. clean lake build;
2. kernel acceptance of every theorem body;
3. #print axioms output for all top-level declarations;
4. clean-environment independent reproduction;
5. hash equality between two builds.

The precise disposition is therefore **partially closed**, not closed. The gap has been narrowed to a mechanical execution: install the pinned toolchain, run lake build, preserve stdout/stderr, and append the axiom report. Any elaboration failure must be repaired in the source and rerun; it may not be converted into documentary proof closure.

Gate 3.5 (Lean kernel acceptance). The Lean gate becomes closed-positive only if all modules build under the pinned toolchain, no release theorem depends on sorryAx or an unregistered custom axiom, and a second clean environment reproduces the dependency report.

4 v15 Gate Zero audit

4.1 Audit question

The Gate Zero question is not whether the v15 prose accurately repeats the v14 tables. It is whether the inherited v8 numerical results can be regenerated from one frozen transition function, source events, registry, configuration, and conformance suite while preserving the v15 definitions and warrant boundaries.

4.2 Artifact inventory

The audit searched the available package for the required executable lineage. Table 3 reports the result.

Table 3: Gate Zero artifact-presence audit.

Required artifact	Available
v8 generator source	No
v8 event log	No
v8 trajectory table	No
v8 registry	No
v8 frozen configuration	No
v8 conformance suite	No
v8 environment lock	No
v8 source figures/source data	No
v8 checksums	No
v15 Article source	Yes
v15 Supplement source	Yes
v15 revision plan	Yes

4.3 Documentary crosswalk

The audit also compared the retained numerical claims in Article and Supplement v15. The following quantities match at the printed precision: 12,800 trajectories, 102,400 events, 23 target-pair witnesses, full-CNS accuracy 0.585, matched-comparator accuracy 0.400, full-CNS certified reach 0.435, rounded best non-CNS certified reach 0.003, and reuse 0.347.

Four article-level values are not reproduced in the compressed v15 Supplement text: the exact 0.0033 comparator reach, containment 0.987, containment 0.717, and master seed 20260715. Some are present in earlier versions, but the source events are absent. The 0.0033 versus 0.003 difference may be ordinary rounding; it cannot be regenerated or verified at exact precision without the event table.

4.4 Gate disposition

Proposition 4.1 (Gate Zero result). *The v15 Gate Zero audit has disposition **closed-negative**. Its machine-readable audit code is*

FAIL_CLOSED_MISSING_EXECUTABLE_LINEAGE.

This closes the audit procedure but blocks positive conformance. The manuscripts may retain the earlier numbers only as historical constructive-witness statements attributed to the earlier execution. They should not call those numbers v15-audited, and they should not use them to promote any v15 semantic, Lean, recurrence, receiver, or external-mapping claim.

4.5 Exact condition for positive reopening

A positive Gate Zero rerun requires the generator, raw append-only event log, trajectory table, canonical registry, frozen configuration, conformance tests, environment lock, figure data, and checksums. The rerun must regenerate every table and figure from events, compare hashes, and classify each inherited result as conformant, partially conformant, nonconformant, or undetermined under v15 definitions.

5 Direct recurrence experiment

5.1 Formal model

Let $U_t \geq 0$ be unresolved target-relevant obligation backlog, A_t arrivals, and S_t certified integration service:

$$U_{t+1} = \max\{0, U_t + A_t - S_t\}.$$

A valid order lift at time T_L reverses the current regime when, over a nonzero interval, service exceeds arrivals and the backlog decreases. Recurrence occurs when continued post-lift growth creates a persistent positive arrival-service gap at now-fixed order $N + 1$.

Theorem 5.1 (Deterministic recurrence criterion). *Suppose there are times $T_L < T_R$ such that:*

1. $U_t > 0$ and $S_t - A_t \geq \delta_1 > 0$ for $T_L \leq t < T_R$, so the current backlog decreases;
2. there is $T_* > T_R$ and $\delta_2 > 0$ such that $A_t - S_t \geq \delta_2$ for all $t \geq T_*$.

Then the lift reverses the backlog over the first interval, and U_t grows without a uniform bound after T_ .*

Proof. While $U_t > 0$ in the reversal interval, $U_{t+1} \leq U_t - \delta_1$, so U_t decreases until it reaches the reflected boundary or the interval ends. After T_* , $U_{t+1} \geq U_t + \delta_2$; induction gives $U_{T_*+n} \geq U_{T_*} + n\delta_2$, which is unbounded in n . $\square$

This theorem separates recurrence from its empirical instantiation. The finite experiment tests whether the registered stochastic regimes produce the predicted paths.

5.2 Frozen protocol

The executable protocol used master seed 20260716, 1,000 replicates per regime, 12 regions, 300 time steps, lift time $T_L = 80$, operand-expansion time $T_E = 120$, and optional second-lift time 215. Before the lift, per-region arrival intensity exceeded service. The lift increased service above arrivals, providing a genuine reversal interval. Continued operand expansion then raised the arrival intensity.

Five regimes were compared:

1. **fixed post-lift service:** lift capacity remains fixed while arrivals continue to grow;

2. **proportional same-order service**: service tracks each region's arrivals with a positive margin;
3. **regenerative distribution**: certified service increases through delayed logistic spread of the new composition;
4. **second order lift**: service receives a second discrete increase at time 215;
5. **branch heterogeneity**: region-specific arrival multipliers create uneven burdens under fixed post-lift service.

Regional fragmentation was the number of regions with backlog at least 12. The primary recurrence endpoint required all of the following: a post-lift decrease of at least 30% relative to the pre-lift backlog; a late aggregate slope above 1 backlog unit per time step; final backlog above 150; and a 30-step window after time 190 satisfying the slope and backlog thresholds. These conditions prevent a noisy positive local slope from being called recurrence when the residual remains small and bounded.

5.3 Results

Table 4: Direct recurrence results across 1,000 replicates per regime. Bracketed values are empirical 2.5th and 97.5th percentiles. "Regions" is the final number of regions above the preregistered fragmentation threshold.

Regime	Rev.	Rec.	Recurrence time	Final backlog	Regions
Fixed post-lift	1.000	1.000	221.8 [219, 236]	625.7 [496, 767]	11.84 [11, 12]
Proportional service	1.000	0.000	-	39.9 [13, 79]	0.72 [0, 3]
Regenerative distribution	1.000	0.000	-	45.2 [17, 82]	0.92 [0, 3]
Second order lift	1.000	0.000	-	49.4 [20, 86]	1.11 [0, 3]
Branch heterogeneity	1.000	0.999	219.3 [219, 225]	918.6 [670, 1200]	9.89 [8, 12]

All five regimes showed the required immediate reversal: mean aggregate slope changed from approximately +6.94 before the lift to between -7.07 and -10.14 during the reversal interval. Recurrence was effectively certain when post-lift service remained fixed and when heterogeneous branches contained persistently under-served regions. It did not satisfy the preregistered persistent-recurrence endpoint under proportional service, regenerative distribution, or the second lift.

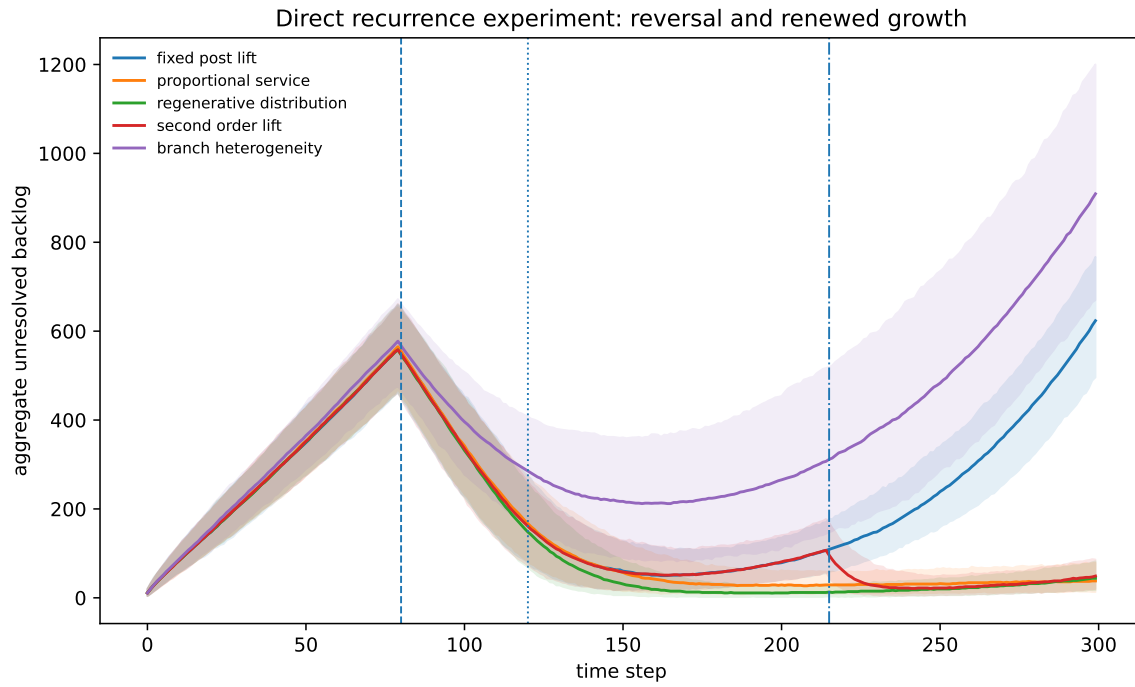


Figure 1: Aggregate unresolved backlog. The dashed line marks the order lift, the dotted line operand expansion, and the dash-dot line the optional second lift. Bands are empirical 95% simulation intervals.

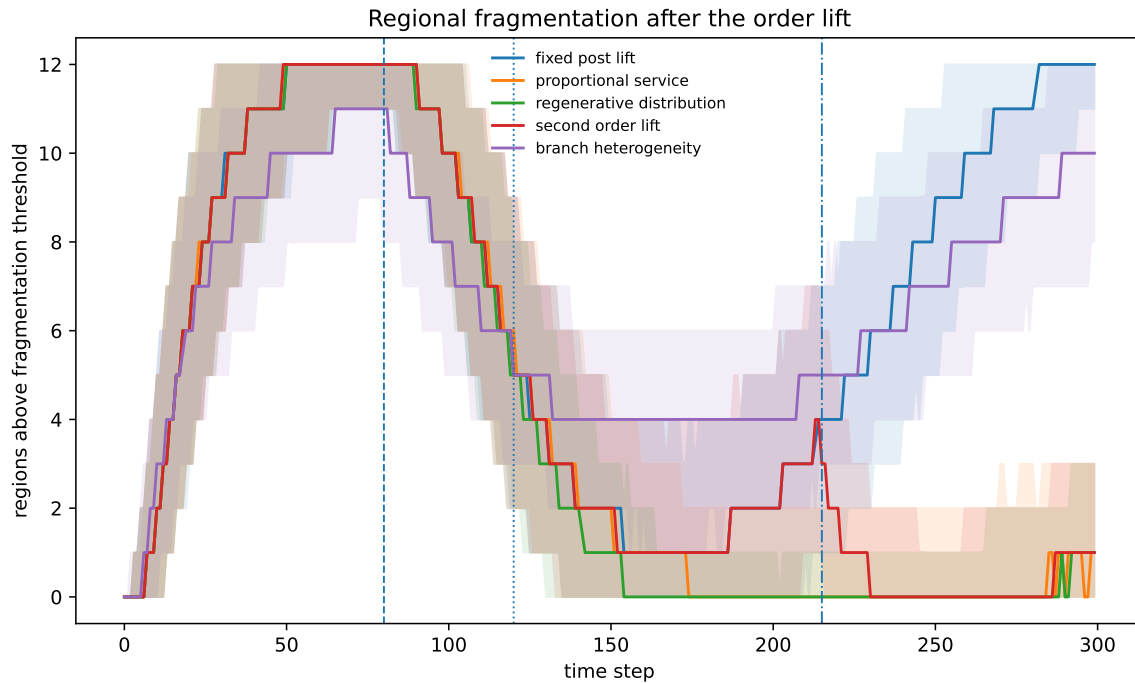


Figure 2: Number of regions above the fragmentation threshold. Fixed post-lift service and heterogeneous branches develop widespread regional fragmentation; scaling and additional lifts keep the median regional count low.

5.4 What this closes

The experiment closes the v15 statement that a direct recurrence run was absent. It supplies an executed finite witness for the following model-relative claim:

After a valid order lift reverses an increasing backlog, continued growth at fixed post-lift service can recreate a positive backlog regime and regional fragmentation; proportional service scaling, regenerative distribution, or another order lift are explicit counter-regimes.

It does not show that current AI systems, institutions, or civilizations follow these parameter values. It also does not audit or replace the inherited v8 realization. Its warrant is *executed synthetic recurrence experiment*.

6 Independent receiver propagation

6.1 Completed protocol objects

The companion package freezes:

- six experimental conditions, from prose-only through CST-plus-Lean and corrupted controls;
- seven primary transfer tasks;
- eight primary endpoints, including second-receiver regenerative success and author-repair count;
- 14 hidden claim families instantiated as 28 randomized tasks;
- a separately stored answer key;
- a scoring schema with fatal-error categories;
- a blinding and no-author-repair rule.

The hidden families cover conflation, reflection without lift, Register/Recall type substitution, span deletion, recurrence countermodels, both CNS-bridge directions, participant completeness, terminal/observability target identity, GF-3 to GF-6 promotion, benign isomorphism, metric/interaction mutation, and external-premise promotion.

6.2 Regenerative endpoint

For receiver generations i and j ,

$$\text{Regenerate}(i, j; \Gamma, T) = 1$$

only if j recovers a contract-preserving representation of Γ , executes it on a hidden target family T , and produces a retransmission package from which a second receiver succeeds without author repair. Mere agreement with the manuscript conclusion does not satisfy the endpoint.

6.3 Why the empirical gate remains open

No fresh human, model, or hybrid receiver was available for recruitment in this execution environment. Running the answer key through a checker written from the same specification would measure conformance to the generator, not independent semantic transfer, and would recreate the tuned-checker problem identified in the v15 plan. The protocol is therefore operationally complete but empirically open.

A prior discussion with another model can motivate task design, but it was not preregistered, blinded, hidden-target, or regenerative and cannot be retroactively converted into the primary study.

Gate 6.1 (Receiver closure). The receiver gate becomes closed only after independent receiver families complete the hidden tasks, a second generation attempts retransmission, author interventions are logged, and analysis treats receiver and claim family—not individual prompts—as inferential units.

7 External mappings

7.1 Registration closure

The external-mapping matrix now gives each mapping a stable identity, formal object, minimum evidence, strongest permitted claim, and explicit defeater. This closes a major documentary gap: no external bridge can be silently promoted merely because its formal premise is convenient.

Table 5: External-mapping certificate summary.

Mapping	Minimum evidence	Status	Strongest permitted statement
Cross-FSS semantic principle	Target-specific invariant-preserving map	Formal hypothesis	The same structural principle can be instantiated across FSSs.
Strong interaction $\leftrightarrow$ Register	Support factorization and schedule-commutation witnesses	Open	Candidate physical realization.
Weak interaction $\leftrightarrow$ Recall	Support nonfactorization and precedence witnesses	Open	Candidate physical realization.
Electromagnetism $\leftrightarrow L$	Support factorization and precedence witnesses	Open	Candidate physical realization.
Gravity $\leftrightarrow G$	Support nonfactorization and commutation witnesses	Open	Candidate physical realization.
Current AI trace mapping	External, version-locked collapsed-pair traces with hidden targets	Open	Testable external mapping.
Historical order lifts	Preregistered uniform classification and independent evidence	Open	Candidate historical instances.
Great Filter branch closure	Heterogeneous branch evidence or defensible external model	Open	Conditional branch premise.
Hazard non-summability	Longitudinal hazard and correction/dispersal evidence	Open	Conditional hazard premise.
Target residue suppression	Outcome- and timescale-specific residue evidence	Open	Conditional outcome premise.

7.2 Why external instantiation cannot be closed here

The physical operation assignments are not standard measurements that can be read directly from established interaction ranges. Their support and schedule definitions are target-relative functional claims requiring explicit factorization, nonfactorization, commutation, and precedence certificates. Current-AI, historical, branch, hazard, and residue mappings similarly require data not contained in the manuscripts or current artifact set.

Accordingly, the package closes *mapping registration* but not *mapping truth*. This distinction is substantive: a mapping can now fail at a named certificate without

affecting the conceptual theorem package, while a successful certificate can promote only the claim level it supports.

8 Integrated status and manuscript consequences

8.1 Updated status tuple

For the five gates addressed here, define

$$\text{Status}_{\text{gates}} = (s_{\text{Lean}}, s_{\text{G0}}, s_{\text{Rec}}, s_{\text{Recv}}, s_{\text{Ext}}).$$

The result of this closure attempt is

$$\text{Status}_{\text{gates}} = (\text{partial}, \text{closed-negative}, \text{closed}, \text{partial}, \text{partial}).$$

This is more informative than an aggregate “open” label. It identifies one new positive result, one definitive carrier failure, and three packages whose construction is complete up to an unavailable execution or external input.

8.2 Required Article changes

The Article should make the following changes before submission:

1. Replace “a direct longitudinal recurrence experiment remains registered” with a concise report of the executed five-regime recurrence result, explicitly labeled synthetic and model-relative.
2. Replace any implication that Gate Zero is merely pending with the negative audit result: the v8 executable lineage is absent from the available submission package.
3. Retain the inherited v8 numbers only as earlier historical constructive-witness results unless the source lineage is restored and regenerated.
4. Describe the Lean artifact as source-complete but not kernel-checked until lake build and the axiom audit actually run.
5. Keep receiver propagation and external mappings explicitly open.

8.3 Required Supplement changes

The Supplement should add:

1. the direct recurrence protocol, theorem, table, and figures from this note;
2. the Gate Zero artifact inventory and fail-closed disposition;
3. the Lean source-tree manifest and static-audit result, with an explicit statement that a lexical scan is not a kernel proof;
4. the frozen receiver protocol and task-family table;
5. the external mapping certificate matrix;
6. the updated gate status tuple.

8.4 Submission consequence

The current package is not yet ready to claim Lean-checked global coherence, independent receiver propagation, or external physical/civilizational instantiation. The direct recurrence result can be reported. The Gate Zero result creates a stronger requirement than before: either supply the v8 executable lineage or narrow/remove the inherited numerical prominence. A clean Lean build plus one independent receiver or external anchor remains the minimum credible path to the v15 carrier envisioned by the revision plan.

9 Exact remaining work order

9.1 Lean

1. Install the pinned Lean 4.32.0 toolchain.
2. Run `lake build` in the companion `lean/` directory.
3. Repair only genuine elaboration/type errors; do not weaken theorem statements silently.
4. preserve `#print axioms` output for all top-level theorems;
5. rerun in a second clean environment and compare hashes.

9.2 Gate Zero

1. Supply the nine missing v8 artifact classes.
2. Rebuild every inherited metric from the raw events.
3. Verify one-generator semantics, label independence, state correspondence, residual propagation, and target-revision separation.
4. issue a per-result conformance disposition.

9.3 Receiver study

1. Freeze the supplied task bank and answer key.
2. Recruit at least three independent receiver families in fresh contexts.
3. randomize conditions and keep the answer key inaccessible;
4. run second-receiver retransmission;
5. analyze receiver-family and claim-family effects.

9.4 External mappings

Execute only the mapping studies whose target claims are intended for the title or abstract. A failed physical-force mapping should narrow that cross-FSS hypothesis, not be hidden and not be allowed to contaminate the conceptual CNS theorem. Great Fil-

ter promotion beyond GF-3 requires branch, hazard, timescale, outcome, and residue evidence rather than additional formal elaboration alone.

10 Conclusion

This note closes what can be closed without fabricating missing evidence. Direct recurrence is now an executed finite result. Gate Zero is no longer vaguely pending: it has a determinate negative disposition because the computational lineage is absent. The Lean formalization, receiver study, and external mappings are substantially more executable than before, but their decisive gates remain unrun or empirical. This is not a failure of completeness. It is the completion of the status record: every claim now has an artifact, an executed result, or a named blocker, and none of the open bridges is disguised as proof.

A Lean project tree and build commands

```
lean/  
  lean-toolchain  
  lakefile.toml  
  FmiCns.lean  
  FmiCns/  
    Basic.lean  
    Semantics.lean  
    Projection.lean  
    Operations.lean  
    Order.lean  
    Backlog.lean  
    GreatFilter.lean
```

```
cd lean  
lake build  
lake env lean FmiCns.lean > axiom-report.txt 2>&1  
grep -R -n -E '\bsorry\b|sorryAx|\baxiom\b' .
```

B Recurrence protocol constants

Parameter	Value
Master seed	20260716
Replicates per regime	1000
Regions	12
Horizon	300
Lift time	80
Operand expansion time	120
Second-lift time	215
Regional backlog threshold	12
Recurrence window	30
Persistent slope threshold	1.0
Final backlog threshold	150

The exact executable protocol is stored in `data/direct_recurrence_protocol.json`. Per-replicate metrics are stored in `data/direct_recurrence_metrics.csv`; compact time-series quantiles are in `data/direct_recurrence_timeseries_summary.csv`.

C Gate Zero audit identifiers

Audit ID: EXP-GATE-ZERO-V15-DOCUMENTARY-20260716
Scope: available conversation/container artifacts only
Result: FAIL_CLOSED_MISSING_EXECUTABLE_LINEAGE

The audit’s JSON result, artifact manifest, claim crosswalk, and available-file hashes are in `audit/`.

D Receiver hidden-family inventory

Family	Target distinction
HF-CONFLATE	collapsed registration with target-different latent states reflective observability versus order lift boundary Register versus conceptual L/G endomorphism
HF-REFLECTION	
HF-REGISTER	
HF-RECALL	memory Recall versus conceptual L/G endomorphism
HF-ORDER	novelty, span, reuse, and non-equivalence
HF-RECURRENCE	fixed service recurrence versus scaled-service counter-model
HF-BRIDGE-LIFT-DYN	order lift plus contraction/closure/invariance

HF-BRIDGE-DYN-LIFT	new dynamical operator requiring a prior-unavailable composition
HF-GOVERNANCE	participant completeness versus aggregate mean
HF-GF-TARGET	terminal versus observability target
HF-GF-PROMOTION	GF-3 conditional result versus GF-6 explanation
HF-SEM-RENAME	benign admissible isomorphism
HF-METRIC	label-preserving target-relevant metric mutation
HF-EXTERNAL	formal proof versus unmeasured external premise

E Companion artifact integrity

The companion archive contains a machine-readable SHA-256 manifest for every file. The archive is designed so that the note can be read independently, while all executed and proposed procedures can be inspected and rerun without recovering definitions from earlier unpublished artifacts.

References

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